

1. Let $x_1, x_2, x_3, \dots$ be an infinite sequence of binary random variables.
 - (a) [2 marks] What does it mean to say that these variables are exchangeable?
 - (b) [3 marks] What form does de Finetti's theorem give for the joint distribution $p(x_1, x_2, \dots, x_n)$?
 - (c) [3 marks] Write down the first three of the Savage axioms, in the form given by de Groot.
 - (d) [6 marks] John elicits a separate prior $p_n(x_1, x_2, \dots, x_n)$ for each value of $n = 1, 2, \dots$. John's priors are not marginally consistent, that is

$$p_n(x_1, \dots, x_n) \neq p_{n+1}(x_1, \dots, x_n, 0) + p_{n+1}(x_1, \dots, x_n, 1).$$

- (i) Show that, under John's priors, the variables $x_1, x_2, x_3, \dots$ do not form an infinite exchangeable sequence.
 - (ii) Show that John's priors do not satisfy the Savage axioms.
- (e) [6 marks] Jane elicits a process generating $x_1, x_2, x_3, \dots$ in which $x_1 = 1$ with probability p and for $n = 1, 2, \dots$,

$$p(x_{n+1} = 1 | x_1, \dots, x_n) = \frac{p + k_n}{1 + n},$$

where $k_n = \sum_{i=1}^n x_i$. Show that this process generates an infinite exchangeable sequence.

2. Bees visit a flower at times $T_0, T_1, \dots, T_n$ with n fixed. Let $y_i = T_i - T_{i-1}, i = 1, 2, \dots, n$. Are these intervals exponentially distributed? In model $m = 1$, $y_i \sim \text{Exp}(\beta)$, with unknown rate $\beta > 0$. Bees may avoid a flower already occupied. In model $m = 2$ the intervals $y_i \sim \text{Gamma}(\alpha, \beta)$ have a Gamma distribution with unknown $\alpha > 1$ and $\beta > 0$. Experience in other studies shows μ is a central value for the interval distribution, and intervals shorter than τ and longer than τ' are improbable.
 - (a) [4 marks] Briefly outline the factors to consider in eliciting priors for the model index m and the model parameters α and β . Why is $\alpha > 1$ in the second model?
 - (b) [3 marks] Let $\theta = \beta$ if $m = 1$ and let $\theta = (\alpha, \beta)$ if $m = 2$. Suppose m and θ have the joint prior $\pi(\theta, m) = \pi_M(m)\pi(\theta|m)$. Write down the joint posterior distribution $\pi(\theta, m|y)$ giving as much detail as you can, but without further prior elicitation.
 - (c) [3 marks] How would you use Monte Carlo samples $\theta^{(t)}, m^{(t)} \sim \pi(\theta, m|y)$, $t = 1, 2, \dots, T$ from the joint distribution to test model 1 against model 2?
 - (d) [7 marks] Give a reversible jump MCMC algorithm targeting $\pi(\theta, m|y)$, specifying the proposals and the acceptance probabilities for the updates which increase and decrease dimension and for a fixed dimension update. You may assume the marginal priors $\pi(\alpha|m = 2)$ and $\pi(\beta|m = 1, 2)$ are easily sampled.
 - (e) [4 marks] Suppose we have $\theta_1^{(t)} \sim \pi(\theta|y, m = 1)$ and $\theta_2^{(t)} \sim \pi(\theta|y, m = 2)$ for $t = 1, 2, \dots, T$. Give an estimator for $\pi(m = 1|y)$ in terms of these samples and show how it may be used to generate samples $y^{(t)}, t = 1, 2, \dots, T$ distributed approximately according to the posterior predictive distribution, $y^{(t)} \sim p(\cdot|y)$ averaged over models. Assume the likelihood $p(y|\theta, m)$ is easily sampled.